

Connor Savugot

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SUMMARY

North Carolina State University | B.S. Computer Engineering | May 2026

Embedded firmware engineer focused on power electronics, BMS integration, communication buses, Embedded Linux, board bring-up, and system-level validation.

TECHNICAL SKILLS

Languages: C, C++, Python, Bash, MATLAB, VBA **Platforms:** MSP430, TMS320, TI AM62/ARM Linux, STM32, dsPIC, FreeRTOS

Systems & Interfaces: Embedded Linux, systemd, IPMI, external watchdogs, BMS, Yocto/Arago, device trees, CAN/CAN FD, UART/RS232, SPI, I2C, ADC, PWM, GPIO, JTAG

Hardware & Tools: Through-hole/SMT soldering, PCB rework, breadboards, KiCad, Git/GitLab, oscilloscopes, logic analyzers

PROFESSIONAL EXPERIENCE

AEGIS Power Systems | Embedded Firmware Engineer

Current

- Developed firmware control for custom bidirectional buck-boost power supplies using 24 independently configurable PWM channels across transformer-coupled input and output stages.
- Built a centralized framework translating requested input/output voltages into coordinated PWM timing, duty cycles, switching behavior, ADC feedback, and safe-operating constraints across paired converter channels.
- Evaluated BMS, monitoring, interface, and support ICs against electrical limits, host protocols, protection, footprints, tool support, and sourcing risk; converted datasheet and reference-design findings into integration plans.
- Root-caused I2C, CAN/CAN FD, SPI, and UART faults by correlating register transactions and analyzer captures with schematics, timing, bus configuration, and known-good hardware; created focused peripheral bring-up tests.
- Characterized IPMI 2.0 and chassis-management services, validating IPMB addressing, sensors, and FRU access while separating transport, session, authentication, and firmware-compatibility failures.
- Created watchdog/systemd services, device-tree and Yocto/Arago support, and Python/Bash tooling for reproducible monitoring, flashing, validation, and field diagnostics on TI AM62-class Linux systems.

AEGIS Power Systems | Embedded Systems Intern

May–Aug 2024 & May–Aug 2025

- Built a secured chroot and restricted SSH CLI on a stripped TI Yocto/Arago system, using custom Bash/Python commands to inspect and control embedded power hardware.
- Created Python/Bash firmware-flashing utilities with packetized transfers, checksum validation, and SPI reflashing for TI TMS320 targets, turning manual update steps into a repeatable workflow.
- Reverse-engineered RS232 command transactions from limited vendor documentation, generated checksums, and helped construct a relay, resistive-load, and power-supply circuit for burn-in monitoring.
- Supported FreeRTOS firmware, ADC/PWM and serial-interface testing, schematic review, and Code Composer Studio debugging; documented Git, firmware-delivery, and toolchain procedures for engineering handoffs.

TEAM Industries | Engineering Intern

Jun–Aug 2021 & Jun–Aug 2022

- Automated quality-control reporting and revision tracking with Excel/VBA, including identification and reporting of out-of-spec production parts.
- Built structured engineering worksheets, custom formulas, and revision records that made manufacturing and quality data easier to review and maintain.
- Collaborated with engineering and manufacturing personnel to document design changes and produce technical reports supporting process and revision reviews.

SENIOR DESIGN

EV Active Sensor Adapter | MSP430FR2355

NC State University

- Defined system architecture and interface specifications integrating an MSP430, BMS/power-management hardware, sensors, custom PCBs, embedded firmware, and telemetry.
- Evaluated BMS components against electrical, sensing, protection, communication, packaging, and sourcing requirements; documented trade studies and translated requirements into hardware/firmware interfaces.
- Prototyped interrupt-driven SPI register transactions and integrated ADC, UART, timers, GPIO, and LCD telemetry, validating each subsystem before full-system integration.
- Maintained requirements, architecture diagrams, interface definitions, test plans, design decisions, and progress reports; coordinated team work and communicated specifications, risks, and results in design reviews.

TECHNICAL LEADERSHIP

Lead Instructor | Project-Based Electronics Workshop

Two Semesters

- Co-led students from project concept through final demonstration; developed and taught through-hole/SMT soldering theory, safety, and hands-on technique using custom NCSU IEEE PCB keepsakes.

Embedded Systems Teaching Assistant

NC State University

- Delivered supplemental lectures and test preparation; taught theory, code debugging, lab techniques, hardware bring-up, project strategy, and ECE advising.
- Used question-driven coaching to build independent troubleshooting skills and lasting professional relationships, raising the class average from its typical C+ to an A-.